SOFTWARE QUALITY ASSURANCE

WHAT IS SQA

Software Quality Assurance (SQA) is simply a way to assure quality in the software. It is the set of activities that ensure processes, procedures as well as standards are suitable for the project and implemented correctly. It is a process that works parallel to Software Development. It focuses on improving the process of development of software so that problems can be prevented before they become major issues. Software Quality Assurance is a kind of Umbrella activity that is applied throughout the software process.

WHY ISO 25010 MATTERS

By understanding and applying these 8 characteristics, developers can:

- Build high-quality and user-friendly software.
- Detect and fix problems before release.
- Ensure the system meets user expectations and industry standards.

THE 8 MAIN QUALITY CHARACTERISTICS

Functional Suitability:

👉 Does the software do what it’s supposed to do?

This checks whether the software performs the right tasks correctly.

THE 8 MAIN QUALITY CHARACTERISTICS

Performance Efficiency

👉 Is it fast and responsive?

This measures how well the software uses computer resources like memory, CPU, and internet speed.

THE 8 MAIN QUALITY CHARACTERISTICS

Compatibility

👉 Can it work with other systems or devices?

This means your software can run on different platforms or communicate with other programs.

THE 8 MAIN QUALITY CHARACTERISTICS

Usability

👉 Is it easy for users to learn and use?

Even if software is powerful, users won’t like it if it’s confusing.
Usability ensures the system is user-friendly and visually clear.

THE 8 MAIN QUALITY CHARACTERISTICS

Reliability

👉 Can it work properly for a long time without crashing?

Reliable software keeps running smoothly without frequent errors or downtime.

THE 8 MAIN QUALITY CHARACTERISTICS

Security

👉 Is the system safe from hackers or unauthorized users?

Security ensures that data is protected and only authorized users can access it.

THE 8 MAIN QUALITY CHARACTERISTICS

Maintainability

👉 Is it easy to fix or improve the code later on?

Good software should be well-organized, so developers can update or fix bugs without breaking other parts.

THE 8 MAIN QUALITY CHARACTERISTICS

Portability

👉 Can it be used in different environments or devices easily?

This checks if the software can move or adapt to another operating system or hardware.

THANK YOU